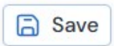
Main Ethics Application Form

Guidance

Before you start

Note: Below is some helpful guidance completing this form. The form accepts plain text only (no special formatting). You can upload attachments to the form if special formatting is required (e.g. charts, illustrations etc.)



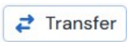
Please note that the session will time out after a period of inactivity. It is advised that you regularly **Save** to ensure no content is lost.

You can use the **Previous** or **Next** buttons to move throughout the form, or use the **Navigate** button to return to the navigation page. Begin from the **START HERE** button, to ensure the correct questions will appear on your form.

To share access to your form, use the **Roles** function, and you can also assign roles through **Assign Role** next to any of the contact boxes on the form. Use the **Collaborators** button to see their level of access

Use the **Completeness Check** to ensure that you have answered all of the relevant questions. Please note you will not be able to submit an incomplete form.

All student projects will require a supervisors signature. You can see a list of signatures here and any pending signature requests. You can **Unlock** an application once signed, but this will require a new signature to confirm changes.



You can **Transfer** your project to another researcher. Please note you will lose access to the project once this is complete.

You can **Unlock for Amendment** on any approved project. Please ensure you change your answer to Question 2. Failure to do so will result in delays.

Screening Tool for Researchers

1.0 Does your project involve any of the following?

- [x] Research involving human participants.
- [x] Access to, collection or use of personal data or property, including mass data collected online (including from social media platforms).
- [ ] Access to, collection of or use of non-personal sensitive or confidential data.
- [ ] Research with the potential to expose any person, whether participating in the research or not, to physical or psychological harm.
- [ ] Research with the potential to cause a significant negative impact or damage to the environment.
- [ ] Research involving genetic material and the local or traditional knowledge relating to the genetic material.
- [ ] Research exploring or involving illegal activities, requiring access to or handling of materials related to illegal activities and/or research that could lead to the disclosure of information that could facilitate illegal activities.
- [ ] The collection and/or use of material derived from humans.

External approval screening

- [ ] Research involving His Majesty's Prison and Probation Services.
- [ ] Research requiring sponsorship and external approval from the Health Research Authority and/or NHS Research Ethics Committee.
- [ ] Research requiring external approvals from the Ministry of Defence Research Ethics Committee.
- [ ] Research involving animals, including both research covered by the Animals (Scientific Procedures) Act 1986 (ASPA) and non ASPA research involving animals and when relevant frameworks exist, to include research involving material derived from live or deceased animals.

Please click the below if none of the above from BOTH checklists apply

- [ ] None of the above.

New Application or Amendment

2.0. Initial or Amendment

- ( ) New application or revision to initial application
- (•) Amendment to an already approved application
- ( ) Amendment to an already approved application from Sussex Direct (Legacy System)

Please ensure that if your application has been unlocked for an amendment, that this above answer has been changed. This will ensure you are revealing all relevant questions on the form. Failing to do so will cause delays.

Amendment Guidance

F-REC's will decide on whether a re-submission constitutes a substantial amendment and requires full ethical review. PI's should seek guidance from the SREO's or Senior Research Ethics and Integrity Officers if in doubt.

F-REC's must be notified of all changes before they take place, no changes may be implemented without approval.

Note that if you intend to investigate a new research question, this will require a new application submission rather than an amendment.

Applicant Details

2.1. Project Title | **JRA**: "Role of the Cardiac Cycle in the Perception of Reality and Illusion"

Applicant

Title	Mr
First Name	Oliver
Surname	Collins
Email	oc236@sussex.ac.uk
Job Title	
Status	Undergraduate
Faculty	Faculty of Science, Engineering and Medicine
Department	Psychology

Supervisor

Title	Dr
First Name	Dominique
Surname	Makowski
Email	D.Makowski@sussex.ac.uk

Who is the Principal Investigator for this project?

If you are a student, this may be your supervisor. Please ask your supervisor to complete this question if you are unsure.

- ○ Applicant
- ● Supervisor
- ○ Another Researcher

Worktribe project ID (if applicable)

2.2. Is this project connected to your supervisors existing research or research data? ○ Yes ● No

2.5. Does the project involve any co-researchers from the University of Sussex? ○ Yes ● No

2.13. Is a data sharing agreement or contract required for your project? ○ Yes ● No

Project Overview

3.0. Proposed start date 15/06/2026

3.1. Proposed end date 01/06/2027

3.2. What type of project are you undertaking?

- [x] Research
- [ ] Service Evaluation
- [ ] Audit
- [ ] Impact
- [ ] Professional Practice
- [ ] Knowledge Exchange

3.4. Medical Specific Screening for SEM Faculty

Please tick all that apply, to ensure your application reaches the correct committee for review.

- ☐ Research which requires participants to be under medical supervision
- ☐ Research involving medical interventions, such as the use of imaging techniques/medical devices.
- ☐ Research relating to patient care and/or the way patient care is delivered
- ☐ Research exploring participants experience of health care or experience of health conditions
- ☑ None of the above

3.5 Are you planning to conduct research involving NHS Patients, their tissue or their data, or conduct research which involves NHS Staff or working in the social care setting?

○ Yes

◉ No

3.6 Is this research project only using secondary data sources, with no primary data collection?

This does not apply to datasets collected from social media, which will require full ethical review.

○ Yes

◉ No

Project Details

4.0 **4.0. Description of project.**

4.0.1. Please provide a brief background summary and the context of your project.

This is an amendment to an already-approved ethics application for a Junior Research Associate (JRA) Scheme project: "Role of the Cardiac Cycle in the Perception of Reality and Illusion". This study will follow the standard operating procedures (the risk assessment form is attached in the ethics). For clarity, the previously-approved ethics application stated that the study would take 1 hour 15 minutes in duration, with this amendment now taking the duration up to 1 hour 30 minutes. For the number of additional tasks and questionnaires included in this amendment, the increased study duration may seem relatively low as the study duration had been overestimated in the previously-approved review.

With an explosion of research in recent years suggesting a link between one's physiological state and perception, this proposal will explore the role of the cardiac cycle in visual illusion sensitivity (subjective illusion magnitude).

Upon arrival at the testing space (Human Psychophysiology Lab, 3B19, Pevensey 2), participants will be presented with the Information Sheet and Consent Form. The Information Sheet explains the collection of physiological data via EEG and ECG, the use of silver/silver chloride electrodes, and advises against participation for those with relevant skin sensitivities or allergies. After indicating their consent (once they have checked ALL the boxes), participants will answer demographic and screening questions (relating to gender, education, age, ethnicity, and medical history of cardiovascular conditions). Such demographic information will be collected to assess sample representativeness. Participants with a medical history of cardiovascular conditions will be excluded from taking part in the study due to concerns regarding the segmentation of the cardiac cycle as a result of differences in the electrical activity of the heart, where this exclusion criterion will be made clear in the recruitment materials. Confidentiality will be ensured, with no identifying information collected; data will be stored only after study completion, allowing withdrawal up to that point but not thereafter due to anonymity

After signing their consent, physiological devices will be fitted to participants. Such devices will include the Muse S Athena electroencephalography (EEG) headband, a PLUX Biosignals respiration belt placed around the chest, and the PLUX Biosignals 3-lead electrocardiography (ECG) system (with the reference electrode placed below the right clavicle, and the positive and negative electrodes placed on the left and right 10th ribs, respectively).

The experiment comprises 4 distinct blocks, with 3 of them presented in a counterbalanced order. Participants first start with the resting state part, and then either the Illusion Game part, or cognitive control part, or interoception part first.

An 8-minute Resting State Task (RST) will be used to collect baseline physiological data and allow participants to adapt to the physiological equipment. A subsequent resting state assessment will take place, constituting the Amsterdam Resting-State Questionnaire (ARSQ; Diaz et al., 2014). Then, participants will be administered, in a counterbalanced order, the Illusion Game, as well as two cognitive control tasks (the Simon task and a flanker task), and an interoception battery of tasks and questionnaires.

The Illusion Game (IG; Makowski et al., 2023) is a reaction task in which participants must make perceptual judgements about visual illusion stimuli (e.g., "Which red line is longer?") as quickly and accurately as possible without making errors. There are 3 illusion types (Müller-Lyer, Ebbinghaus, and Vertical-Horizontal), where the illusion strength and task difficulty of the targets are manipulated orthogonally and continuously. This procedure encompasses 2 sets of 80 trials for each illusion. Illusion sensitivity is assessed as the average error rate in the incongruent condition, for each of the 3 illusion types.

The cognitive control tasks will be implemented in a manner similar to Makowski et al. (2020), where two simple experimental neuropsychology tasks will be administered: the Simon task and a flanker task. The order of these tasks will be counterbalanced.

The interoception block will comprise 2 tasks and 3 questionnaires aiming to assess interoceptive abilities: a tapping task, the Heartbeat Counting Task (HCT; Schandry, 1981), the Multidimensional Assessment of Interoceptive Awareness Second Version (MAIA-2; Mehling et al., 2018), the Interoceptive Accuracy Scale (IAS, Murphy et al., 2020), and the Multimodal Interoception Questionnaire (Mint; Makowski et al., 2025).

After completing all tasks, participants will be presented with the electronic debriefing information, in which they will be told the aims of the research project and given the opportunity to ask any questions to the researcher. Physiological sensors will then be removed and participants will be paid 10 GBP in cash.

References:

Diaz, B. A., Van Der Sluis, S., Benjamins, J. S., Stoffers, D., Hardstone, R., Mansvelder, H. D., Van Someren, E. J. W., & Linkenkaer-Hansen, K. (2014). The ARSQ 2.0 reveals age and personality effects on mind-wandering experiences. Frontiers in Psychology, 5. https://doi.org/10.3389/fpsyg.2014.00271

Makowski, D., Sperduti, M., Blondé, P., Nicolas, S., & Piolino, P. (2020). The heart of cognitive control: Cardiac phase modulates processing speed and inhibition. Psychophysiology, 57(3), e13490. https://doi.org/10.1111/psyp.13490

Makowski, D., Te, A. S., Kirk, S., Liang, N. Z., & Chen, S. H. A. (2023). A novel visual illusion paradigm provides evidence for a general factor of illusion sensitivity and personality correlates. Scientific Reports, 13(1), 6594. https://doi.org/10.1038/s41598-023-33148-5

Makowski, D., Neves, A., Benn, E. L., Bennett, M., & Poerio, G. (2025). The Mint Scale: A Fresh Validation of the Multimodal Interoception Questionnaire and Comparison to the MAIA, BPQ and IAS. PsyArXiv. https://doi.org/10.31234/osf.io/8qrht_v1

Mehling, W. E., Acree, M., Stewart, A., Silas, J., & Jones, A. (2018). The Multidimensional Assessment of Interoceptive Awareness, Version 2 (MAIA-2). PLOS ONE, 13(12), e0208034. https://doi.org/10.1371/journal.pone.0208034

Murphy, J., Brewer, R., Plans, D., Khalsa, S. S., Catmur, C., & Bird, G. (2020). Testing the independence of self-reported interoceptive accuracy and attention. Quarterly Journal of Experimental Psychology, 73(1), 115–133. https://doi.org/10.1177/1747021819879826

Schandry, R. (1981). Heart beat perception and emotional experience. Psychophysiology, 18(4), 483-488.

4.0.2. Please provide the research question and/or hypothesis.

This project aims to investigate the role of the cardiac phase (systole versus diastole) on visual illusion sensitivity (the perceived strength of visual illusions).

It is predicted that, among the incongruent trials of the Illusion Game, there will be a greater number of errors in systole relative to diastole.

4.0.3. Please provide a rationale for the overall aims of the project and briefly summarise the expected benefits of your research to your participants or the wider community.

The rationale for this project is to provide evidence supporting a differential role of the two phases of the cardiac cycle (systole and diastole) on the perception of visual illusions (VIs) - specifically the perceived magnitude of a VI (i.e., VI sensitivity). It is currently unknown whether the perception of VIs is associated with the cardiac cycle, with no previous research exploring this relationship.

The cardiac cycle can be divided into two phases: (1) systole, where the heart's ventricles contract, ejecting blood into the arteries, with blood pressure peaking, and (2) diastole, in which the ventricles refill with blood, representing a period of relaxation and minimal blood pressure (Pollock & Makaryus, 2025). Baroreceptors within the arteries detect information pertaining to blood pressure across the cardiac cycle according to the stretching of arterial walls, leading to substantial baroreceptor firing during systole but minimal activity during diastole (Rau & Elbert, 2001). Signals associated with baroreceptor firing, through the glossopharyngeal and vagus nerves, are transmitted to the nucleus tractus solitarius, from where they project to a wide network of brain areas, including the insula, cingulate cortex, and prefrontal cortex (PFC; Suarez-Roca et al., 2021). Within these regions, evidence suggests that systolic baroreceptor projections suppress cortical excitability, leading to attenuated exteroceptive processing and cognition, with conversely more accurate processing during diastole (Skora et al., 2022). Therefore, communication between the cardiovascular and nervous systems is suggested to create interoceptive-exteroceptive interactions influencing perception and cognition.

VIs, creating dissonance between objective reality and phenomenological experience, seemingly provide a useful paradigm to explore the role of these interoceptive-exteroceptive interactions through their theoretical influence on higher-level cognitive processes – an area ostensibly relevant to, but arguably neglected by, existing cardiac phase research. This is where the discrepant experience of viewing a VI may create neural conflict, which the brain aims to resolve through cognitive control mechanisms (Bednarek et al., 2024). Conflict monitoring and cognitive inhibition, associated with the cingulate cortex and PFC – regions receiving baroreceptor projections – may be recruited to minimise such conflict and suppress misleading perceptual information, endeavouring to achieve a more accurate percept, despite the illusion generally persisting. Accordingly, during systole, baroreceptor activation may suppress cortical excitability within the cingulate cortex and PFC, consequently impairing cognitive control mechanisms, leading to enhanced illusion sensitivity.

Successful completion of this project will bolster existing research literature indicating an association between perception, cognition, and the physiological state of the perceiver, demonstrating potentially broad implications for perception. This may be particularly interesting when comparing responses across the IG and cognitive control tasks, where we argue that VIs might elicit psychophysiological responses similar to those of classic cognitive control tasks. This line of research is pertinent to existing projects at the University of Sussex relating to interoception, psychophysiology, psychophysics, consciousness, and reality perception.

References:

Bednarek, H., Przedniczek, M., Wujcik, R., Olszewska, J. M., & Orzechowski, J. (2024). Cognitive training based on human-computer interaction and susceptibility to visual illusions. Reduction of the Ponzo effect through working memory training. International Journal of Human-Computer Studies, 184, 103226. https://doi.org/10.1016/j.ijhcs.2024.103226

Pollock, J. D., & Makaryus, A. N. (2025). Physiology, Cardiac Cycle. In StatPearls. StatPearls Publishing. http://www.ncbi.nlm.nih.gov/books/NBK459327/

Rau, H., & Elbert, T. (2001). Psychophysiology of arterial baroreceptors and the etiology of hypertension. Biological Psychology, 57(1–3), 179–201. https://doi.org/10.1016/S0301-0511(01)00094-1

Skora, L. I., Livermore, J. J. A., & Roelofs, K. (2022). The functional role of cardiac activity in perception and action. Neuroscience & Biobehavioral Reviews, 137, 104655. https://doi.org/10.1016/j.neubiorev.2022.104655

Suarez-Roca, H., Mamoun, N., Sigurdson, M. I., & Maixner, W. (2021). Baroreceptor Modulation of the Cardiovascular System, Pain, Consciousness, and Cognition. Comprehensive Physiology, 11(2), 1373–1423. https://doi.org/10.1002/j.2040-4603.2021.tb00152.x

Makowski, D., Neves, A., Benn, E. L., Bennett, M., & Poerio, G. (2025). The Mint Scale: A Fresh Validation of the Multimodal Interoception Questionnaire and Comparison to the MAIA, BPQ and IAS. PsyArXiv. https://doi.org/10.31234/osf.io/8qrht_v1

4.1. Which research methods do you plan to use?

Please tick all that apply

- [ ] Interviews
- [x] Questionnaires
- [ ] Observation
- [ ] Focus Group
- [ ] Ethnographic methods
- [ ] Citizen Science
- [ ] Workshops
- [x] Experimental
- [ ] Participatory Methods
- [ ] Another Research Method not listed

4.1.1. Please add details on your method/s here

Participants will complete an eight-minute Resting State Task (RST) to collect baseline physiological data and allow them to adapt to the physiological equipment, in which they are told to relax and remain seated quietly, with their eyes closed, but without falling asleep. A subsequent resting state assessment will take place, constituting the Amsterdam Resting-State Questionnaire (ARSQ; Diaz et al., 2014), assessing mind-wandering along seven dimensions: Discontinuity of Mind, Theory of Mind, Self, Planning, Sleepiness, Comfort, and Somatic Awareness. The questionnaire comprises 21 items rated on 7-point Likert scale from 0 ("Completely Disagree") to 6 ("Completely Agree").

Participants will complete the Illusion Game (IG; Makowski et al., 2023), a validated visual illusion sensitivity task, in which they must make perceptual judgements across three visual illusion types (Ebbinghaus, Muller-Lyer, Vertical-Horizontal). Here, participants will be faced with 2 sets of 80 trials for each illusion type, with two illusions (of the same type) presented on the screen simultaneously, where the strength of these illusions is set according to orthogonally manipulated parameters (task difficulty and illusion strength). Participants must answer a two-alterative forced choice question for each trial, asking them which component of each illusion is bigger/longer (for example, in the Vertical-Horizontal illusion trials, participants are asked, in reference to the two simultaneously-presented illusions, "Which red line is longer?"). Participants are told that they should equally aim to maximise their speed and accuracy across all trials, with an arbitrary composite score shown after the blocks for each illusion type as a mechanism for gamification. A practice set of 18 trials (6 per illusion type) will be completed before the 480 experimental trials. Other research projects utilising this identical IG task have received ethical approval previously (e.g., Review Reference: 2026-1527-1475).

The interoception block always begins with the tapping tasks, followed by the questionnaires and the HCT, presented in a counterbalanced order, where these tasks have received ethical approval in a previous study (Review Reference: 0294). The tapping task comprises several subtasks:
1. Voluntary external tapping (40 trials): participants press the spacebar when the rotating clock hand reaches a designated target point (red arrow).
2. Voluntary internal tapping (40 trials): participants press the spacebar voluntarily before the clock hand completes a 360° rotation.
3. Mixed task (80 trials): a randomized combination of the first two subtasks.
4. Rhythmic task: participants synchronize their taps with 10 beeps at 42 BPM and continue tapping at the same rhythm after the beeps stop (120 presses in total).
5. Random tapping (120 presses): participants tap the spacebar at random intervals.
6. Cardiac tapping (120 presses): participants attempt to tap the spacebar in synchrony with their perceived heartbeats.

The Heartbeat Counting Task (HCT; Schandry, 1981) consists of six trials with intervals randomly varying between 20 and 45 seconds. In each trial, participants silently count their heartbeats without physically checking their pulse, then report the number counted and their confidence in its accuracy.

The Multimodal Interoception Questionnaire (Mint; Makowski et al., 2025) is a 33-item questionnaire assessing interoception across modalities and contexts. Each item is rated on 7-point Likert scale (0 = "Disagree", 6 = "Agree").

The MAIA-2 (Mehling et al., 2018) is a 37-item questionnaire assessing eight dimensions of interoception: Noticing, Not-Distracting, Not-Worrying, Attention Regulation, Emotional Awareness, Self Regulation, Body Listening, and Trust. Responses are rated on a 7-point Likert scale from 0 ("Never") to 6 ("Always").

The IAS (Murphy et al., 2020) is a 21-item questionnaire assessing subjective interoceptive accuracy. Items are rated on 5-point Likert scale from 1 ("Disagree strongly") to 5 ("Strongly agree").

Two cognitive control tasks will include the Simon task and a flanker task. The Simon task tests stimulus-response compatibility in a speed-accuracy task (Simon & Rudell, 1967). Participants will view the words "LEFT" or "RIGHT" appearing randomly on either the left or right side of the screen and must respond to the word's meaning by pressing the corresponding key attributed to that stimulus (e.g., F for "LEFT", J for "RIGHT"), while ignoring where the word appears on the screen (as the location of the word stimulus is unrelated to the task). The flanker task (Eriksen & Eriksen, 1974) will consist of five arrows presented on the screen, where participants must respond to the direction of the central arrow (left, right) by pressing the corresponding arrow key on the keyboard.

The surrounding flanking arrows will either be congruent (facing in the same direction as the central arrow) or incongruent (facing in the opposite direction as the central arrow).

References:

Diaz, B. A., Van Der Sluis, S., Benjamins, J. S., Stoffers, D., Hardstone, R., Mansvelder, H. D., Van Someren, E. J. W., & Linkenkaer-Hansen, K. (2014). The ARSQ 2.0 reveals age and personality effects on mind-wandering experiences. Frontiers in Psychology, 5. https://doi.org/10.3389/fpsyg.2014.00271

Eriksen, B. A., & Eriksen, C. W. (1974). Effects of noise letters upon the identification of a target letter in a nonsearch task. Perception & Psychophysics, 16(1), 143–149. https://doi.org/10.3758/BF03203267

Makowski, D., Te, A. S., Kirk, S., Liang, N. Z., & Chen, S. H. A. (2023). A novel visual illusion paradigm provides evidence for a general factor of illusion sensitivity and personality correlates. Scientific Reports, 13(1), 6594. https://doi.org/10.1038/s41598-023-33148-5

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Murphy, J., Brewer, R., Plans, D., Khalsa, S. S., Catmur, C., & Bird, G. (2020). Testing the independence of self-reported interoceptive accuracy and attention. Quarterly Journal of Experimental Psychology, 73(1), 115–133. https://doi.org/10.1177/1747021819879826

Schandry, R. (1981). Heart beat perception and emotional experience. Psychophysiology, 18(4), 483-488.

Simon, J. R., & Rudell, A. P. (1967). Auditory S-R compatibility: The effect of an irrelevant cue on information processing. Journal of Applied Psychology, 51(3), 300–304. https://doi.org/10.1037/h0020586

4.2. Is this research going to be funded by a grant?

(•) Yes ( ) No

4.2.1. Please detail the grant awarding body and status of grant application.

University of Sussex Junior Research Associate (JRA) Scheme - approved

4.3. Will your project involve the utilisation of non-human genetic resources (including any associated traditional knowledge) sourced from outside the UK?

( ) Yes

(•) No

Does your project involve any of the following?

5.0. A physical risk to participants and/or the researcher

e.g. collection of human materials, administration of food, drugs, placebos or other substances, the use, production, storage, waste, transportation and/or release of chemicals and hazardous (biological agents, flammable/dangerous/explosive substances, biological agents etc.) or equipment, the use of physical agents (excessive noise exposure, ionizing radiation, electromagnetic fields) or use of invasive or potentially harmful procedures.

( ) Yes

(•) No

5.1. Potential to induce psychological stress, distress or anxiety, or produce humiliation or cause harm or other negative consequences

- ○ Yes
- ◉ No

5.2. Studies that may lead to disclosures from the participant that raise ethical or moral dilemmas, involvement in illegal actions or risk of harm to themselves or others

- ○ Yes
- ◉ No

5.3. Involvement of participants who could be considered vulnerable or could feel coerced or obliged to take part, or disadvantaged by not taking part.

e.g. people who are unable to give informed consent or in a dependent position, people under 18 years of age, people with learning disabilities, over-researched groups or people in care facilities

- ○ Yes
- ◉ No

5.4. Deception and/or participation without consent (incl covert observation of people in non-public places). Please refer to the British Psychological Society Code of Ethics and Conduct (or similar guidelines) for further information.

- ○ Yes
- ◉ No

5.5. Potential to identify research participants in publications or outputs.

This does not include taking email details for participant prize draws or identifying participants from signed consent forms or holding identity encryption spreadsheets that are stored securely separate from the research data.

- ○ Yes
- ◉ No

5.6. Research exploring or involving illegal activities, requiring access to or handling of materials related to illegal activities and/or could lead to the disclosure of information that could facilitate illegal activities.

- ○ Yes
- ◉ No

5.7. The collection of personal, special categories of personal data*

identifiers relating to racial, or ethnic origin, political opinions, trade union membership, religious or philosophical beliefs, genetics data, biometrics data, health, data concerning sex life or sexual orientation.

- ◉ Yes
- ○ No

5.7.1. Please explain why the collection of any special categories of personal data is necessary for your project and why the project aims cannot be achieved without collection of this data.

A Data Protection Impact Assessment may also be required, so please contact a Data Protection Officer for advice

Data on ethnicity will be collected to assess the study's sample representativeness to understand the generalisability of the findings. However, participants will not be required to answer the questions "How would you describe your ethnicity?" and "In which country are you currently living?", unlike the other demographic questions.

5.8. Please outline any other ethical issues that you think are relevant to your project and not covered elsewhere, including potential conflicts of interest.

N/A

Based upon your answers to the above questions your application may be considered **HIGH** risk.

If, however you wish to make a case that your application should be considered as **LOW** risk please enter the reasons here. Researchers should note that SREOs or F-RECs may decide NOT to agree with the case that you have made.

5.9. Would you like your application to be considered as LOW risk?

- (●) Yes
- ( ) No

5.9.1. Please summarise why:

We consider this study to be low risk as participation poses no harm and does not involve sensitive information or vulnerable populations. Although information is being collected related to ethnicity and sexual sensations, the risk of a participant experiencing harm in this study would be no more than in their day-to-day life, particularly as participants are informed in the Information Sheet the general nature of the questions they will be asked and also do not have to respond to the questions relating to their ethnic background.

Recruitment

6.0. **How many participants do you plan to recruit? Please provide an estimate if unknown at this point.**

If this is a secondary data study, please include the number of data points.

40

6.0.1. Please detail the justification for your sample size.

Approximately 40 participants will be recruited based on the sample sizes of other cardiac phase experiments.

6.1. Please explain how participants will be selected including any inclusion and exclusion criteria.

Participants will sign-up to the study themselves by selecting a convenient timeslot on Calendly, where the Calendly booking page will be accessed through a QR code on the recruitment poster, or through the link in the invitation text. Individuals with diagnosed cardiovascular conditions will be excluded from participating due to possible artefacts in the electrocardiography (ECG) data collection, where this exclusion criterion is stated explicitly in the recruitment materials. Participants must be at least 18 years of age and possess English language fluency.

6.2. Will you require cooperation of a gatekeeper in order to access participants?

- ○ Yes
- ◉ No

6.3 Will you be inviting participants to take part in this research project?

- ◉ Yes
- ○ No

6.3.1 How will initial contact be made with participants?

Please note that if you are recruiting participants via social media, a separate profile should be created (using a university e-mail address or a dedicated study webpage, etc.). Please also review the University's guidance on social media.

Please explain how you will have appropriate access to your participant cohort(s) that is compliant with data protection laws.

Participants will be recruited by responding to a recruitment poster (attached) created specifically for the research project. Participants will be able to book onto a data collection timeslot by accessing the Calendly link using the QR code on the poster. The Calendly link displays all available timeslots, and provides information about what the study will entail, eligibility requirements, and preparation details. The applicant's University of Sussex email address (oc236@sussex.ac.uk) will be provided on the poster and Calendly booking page if participants wish to enquire about the study. The supervisor's email address (D.Makowski@sussex.ac.uk) will also be provided on the Calendly page for any concerns regarding the study.

The recruitment poster will be placed around the University of Sussex campus and will also be uploaded onto my LinkedIn profile (linkedin.com/in/oliver-collins).

Additionally, an invitation message (attached) will be sent via email to colleagues and relatives potentially interested in participating, such as through study swaps.

Data Collection

6.4. Will you be using your University of Sussex or BSMS email address for email correspondence.

- ☑ Yes
- ☐ No

6.5. Please upload copies of the invitation text such as poster, email and web advertisements.

Documents

Type	Document Name	File Name	Version Date	Version	Size
Other Participant Materials	invitation_text	invitation_text.docx	09/06/2026	2	15.9 KB
Other Participant Materials	recruitment_new	recruitment_new.pdf	09/06/2026	2	31.3 MB

6.6 Will you be using any of the following: questionnaires, topic guides, interview structure, debriefs.

- (•) Yes
- ( ) No

6.6.1 **Please upload any participant facing documents related to your intervention, i.e. questionnaires, topic guides, interview structure, debrief.**

Your information sheet and consent form should uploaded later in the form in their respective sections.

Documents

Type	Document Name	File Name	Version Date	Version	Size
Other Participant Materials	debriefing	debriefing.pdf	07/05/2026	1	69.1 KB
Other Participant Materials	ARSQ	ARSQ.pdf	07/05/2026	1	128.0 KB
Other Participant Materials	MINT	MINT.pdf	07/05/2026	1	7.8 MB
Other Participant Materials	demographics	demographics.pdf	07/05/2026	1	71.2 KB
Other Participant Materials	experiment_feedback	experiment_feedback.pdf	07/05/2026	1	35.5 KB
Other Participant Materials	illusiongame_info	illusiongame_info.pdf	07/05/2026	1	76.6 KB
Other Participant Materials	restingstate_instructions	restingstate_instructions.pdf	07/05/2026	1	37.2 KB
Other Participant Materials	cardiovascular_exclusion	cardiovascular_exclusion.pdf	08/05/2026	1	30.7 KB
Other Participant Materials	IAS	IAS.pdf	09/06/2026	1	393.4 KB
Other Participant Materials	MAIA	MAIA.pdf	09/06/2026	1	856.9 KB
Other Participant Materials	interoception_block	interoception_block.pdf	09/06/2026	1	3.7 MB

6.7 Will you be showing participants any photo's/pictures/audio/visual recordings/clips? (•) Yes ( ) No

6.7.1. Please upload any materials which will be shown to participants.

Documents

Type	Document Name	File Name	Version Date	Version	Size
Photo's/Pictures/Audio/Visual Recordings/Clips	ebbinghaus_instructions	ebbinghaus_instructions.pdf	07/05/2026	1	72.7 KB
Photo's/Pictures/Audio/Visual Recordings/Clips	mullerlyer_instructions	mullerlyer_instructions.pdf	07/05/2026	1	63.2 KB
Photo's/Pictures/Audio/Visual Recordings/Clips	verticalhorizontal_instructions	verticalhorizontal_instructions.pdf	07/05/2026	1	56.9 KB

If your research intervention and materials are likely to develop once the research has begun, please include as much information as possible about the planned research intervention and materials you will use.

N/A

6.8 Will participants be provided with a Participant Information Sheet?

- (•) Yes
- ( ) No

6.8.1 Please upload a copy of your Participant Information Sheet.

You are encouraged to utilise the template and guidance notes for research studies:

www.sussex.ac.uk/staff/research/documents/partictipant-information-sheet-template.doc

www.sussex.ac.uk/staff/research/documents/pis-cf-why-do-i.pdf

Documents

Type	Document Name	File Name	Version Date	Version	Size
Participant Information Sheet	information_sheet3	information_sheet3.pdf	09/06/2026	3	158.3 KB

6.9. Are you accessing participants or data containing personal data via an online environment or internet setting?

Please note that when researching over researched or potentially vulnerable groups, it is not appropriate to scrape from online groups which explicitly state that gatekeeper approval is required, or disallow data collection in their group T&Cs.

e.g. chat rooms, social media, instant messaging, online forums.

- ( ) Yes
- (•) No

6.11. Do you intend to collect a large dataset to use for modelling in a machine learning project?

- ( ) Yes
- (•) No

6.12. Does this project involve the use and/or storage of human tissue?

- ○ Yes
- ◉ No

Informed Consent and Withdrawal

7.0. Will participants give consent to take part prior to their participation?

- ◉ Yes
- ○ No

7.0.2. How will your participants give consent?

- [x] In Writing/online form
- [ ] Verbally

7.0.4 **In Writing/online form**

Documents

Type	Document Name	File Name	Version Date	Version	Size
Written Consent Form	consent	consent.pdf	07/05/2026	1	86.6 KB

Consent form templates can be found on our central website

Please find BSMS specific templates on the central BSMS website

7.1. Will questionnaires be completed anonymously and returned indirectly?

- ◉ Yes
- ○ No

7.2. How will you ensure that the participant information is provided in a suitable format for your target group? i.e, do they require language translation, child specific documents, written forms or oral consent?

Participants will be 18+ and fluent in English, so no language translation or child-specific documents will be required. All participant information will be provided in English, in a manner that is easily understandable. Participants will read the consent form and provide confirmation of their consent before beginning the study.

7.3. How long will potential participants have to consider taking part in your study?

Upon arrival at the lab, participants will have as long as necessary to consider taking part before ticking all boxes on the consent form. Participants can also withdraw at any time before the end of the study (duration: approximately 1 hour and 15 minutes) until data is saved.

7.4. Will participants be able to leave at any time during the study without giving a reason?

- (x) Yes
- ( ) No

7.5. Will participants be able to withdraw their research data?

- ( ) Yes
- (x) No

7.5.2. Please explain why participants are unable to withdraw their research data.

This will need to be clearly stated in the participation information sheet or justified if conducting research without consent

> Participants are informed that they may withdraw from the experiment at any time prior to data being saved. However, due to the anonymous nature of participation, once the data has been stored it will no longer be possible to link individual participants to their responses. This will be explicitly stated in the consent form.

7.6. Will participants be reimbursed or paid for their expenses and/or time or be offered to enter a prize draw?

- (x) Yes
- ( ) No

7.6.1. What re-imbursement/payment will participants receive and who is funding this?

> Participants will be paid 10 GBP for taking part in the study (approximately 1 hour and 30 mins in duration), even if they withdraw at any point. This remuneration is being funded using the 3,500 GBP Junior Research Associate (JRA) Scheme bursary.

7.7 Will researchers be publishing direct quotes from research outputs?

- ( ) Yes
- ( ) No
- (x) Not Applicable

Researcher/Participant Safety and Wellbeing

8.0. Is DBS (Disclosure and Barring Service) clearance necessary for this project?

- ( ) Yes
- (x) No

8.1. Will the research be conducted outside the UK? (not including research online)

- ( ) Yes
- (x) No

8.2. Will any local (overseas) ethics or governance approvals or permissions be required in order to conduct the research?

- ○ Yes
- ◉ No

8.3. Please detail all the locations for your research (computer, field work, lab work).

The project will constitute lab work, taking place at the University of Sussex in the Human Psychophysiology Lab (HPL), Pevensey 2, Room 3B19.

8.4. Will participants be provided offered the option of a chaperone?

- ○ Yes
- ◉ No

8.5. Will any researchers be in a lone working situation?

- ○ Yes
- ◉ No

8.6. Will the study involve engaging participants in the discussion of potentially distressing or sensitive topics? (e.g. sexual activity, drug use, ethnicity, political behaviour, potentially illegal activities).

- ○ Yes
- ◉ No

8.7. Can you think of anything else that might be potentially harmful to the research group?

- ○ Yes
- ◉ No

8.8. How will the findings of the study be fed back to participants in an accessible way?

Experiment's goal and hypotheses are described at the end of the experiment. The findings will be disseminated in an open and accessible format. The anonymised dataset and analysis scripts will be made openly available via GitHub and OSF, together with their HTML equivalents so that participants can easily follow each step of the analysis. A summary of the results will be shared in the form of a GitHub blog post and on other platforms (e.g., LinkedIn) to maximise accessibility. Findings will be disseminated at the JRA Poster Exhibition, and potentially at conference events.

Data Storage/Management

9.1. Will data be transferred outside of the University of Sussex?

- ◉ Yes
- ○ No

9.1.1. If anyone outside of University has access to the dataset, you may require data-sharing or contractual agreements. Please confirm you have checked with the Contracts and IP team and this will be in place before the project begins.

- [x] I confirm

9.2. Will you be transferring or receiving any personal data from outside of the UK?

- ( ) Yes
- (•) No

9.3. Will the Principal Investigator take full responsibility during the study, for ensuring the lawful collection of, appropriate storage of and security of information (including research data, consent forms and administrative records)?

- (•) Yes
- ( ) No

9.4. Where will data be stored for the duration of the project?

University of Sussex Box

Data Analysis / Management

9.5. Who will have access to personal information and data relating to this study and how will the results be analysed?

Personal information and data will be available to the research team only: myself (applicant), supervisor (PI), and postgraduate mentor (PI's PhD candidate).

9.5.1 What type of research data will your project be collecting? Please tick all appropriate boxes below:

9.5.1 What type of research data will your project be collecting? Please tick all appropriate boxes below:

- [ ] Paper written notes/field notes
- [ ] Email correspondence
- [ ] Audio recording of participants
- [ ] Audio/video recordings (e.g. Teams or Zoom calls)
- [x] Survey/questionnaire responses (online) e.g. Qualtrics
- [x] Medical data such as scan data or EEG readings
- [x] Other electronic data e.g. diary entries, app data
- [ ] Other

9.5.3 Please detail how you will manage the types of data you are collecting (ticked above) and indicate what safeguards will be in place to ensure that data will be secure and compliant with data protection laws.

Data will be stored in a University of Sussex Box folder with access limited only to the researchers involved in data analysis. Following further anonymisation, the data will be transferred to a private OSF repository (EU servers located in Germany so compliant with GDPR and data protection legislation).

9.6. Will you require the use of transcription software?

- ◯ Yes
- ◉ No

9.7. Is your research data already anonymous?

- ◉ Yes
- ◯ No

9.8. Please detail how long will raw data be retained, and how and when personal data will be deleted?

This includes audio/video files recorded for transcription purposes or media artifacts. Please state if these will be deleted upon transcription, or retained for future research and research outputs (this should be clearly identified in the PIS, consent and media release forms).

> The raw data will be securely stored in a restricted University of Sussex Box folder indefinitely for accountability. Prior to sharing, an additional cleaning procedure will be applied to ensure further anonymisation and remove any potentially identifying information. Only this fully anonymised and processed dataset will be made available publicly via GitHub and OSF, in accordance with best practices for confidentiality and data protection.

9.9. Will lists of identifiable numbers or pseudonyms linked to names and/or other personal information be stored securely and separately from the research data?

- ◉ Yes
- ◯ No

9.10. Do you intend to use the research data for any purpose other than that for which consent is explicitly given?

- ◯ Yes
- ◉ No

9.11 Do you plan to publish identifiable information about participants? (such as including participant names in the write-up)

- ◯ Yes
- ◉ No

9.12 Do you intend to present your research externally? (This might include publication, results in a conference).

- ◉ Yes
- ◯ No

9.12.1. Please specify where:

> JRA Poster Exhibition, social media posts (e.g., LinkedIn, GitHub), possible conference presentations, and potentially in a peer-reviewed publication.

9.13 What will happen to anonymised data from the study after completion of the project? (Please tick all that apply).

- ☐ Data will be transferred to supervisor for future use
- ☑ Data will be used in future research projects
- ☑ Data will be published on an open access repository
- ☐ Data will be destroyed once award is conferred or project is complete
- ☐ Another use

9.14 How will you obtain consent to publish or use the data from this project in future research?

Participants are asked to provide consent via the information and consent sheet, which explicitly states that the data may be used in future research.

9.15 If you have had a change requested through track changes, and you want to discuss this with the F-REC or query this– please use this box to detail your argument

9.16 If you have any other documents to upload which are not specifically related to a question on the form, please attach these here.

Documents

Type	Document Name	File Name	Version Date	Version	Size
SAE Supporting Documentation	devices	devices.pdf	07/05/2026	1	51.5 KB
SAE Supporting Documentation	Risk_Assessment_Form	Risk_Assessment_Form.pdf	07/05/2026	1	2.9 MB

Amendment Details

Amendment Guidance

F-REC's will decide on whether a re-submission constitutes a substantial amendment and requires full ethical review. PI's should seek guidance from the SREO's or Senior Research Ethics and Integrity Officers if in doubt.

F-REC's must be notified of all changes before they take place, no changes may be implemented without approval.

Note that if you intend to investigate a new research question, this will require a new application submission rather than an amendment.

A1.0. Please summarise the changes you are making to this application.

This amendment comprises the addition of a battery of interoception tasks and questionnaires, as detailed in Section 4. As such, the study duration has been increased to 1 hour and 30 minutes. The recruitment poster, invitation text, and information sheet have been updated to reflect this duration change.

A1.1. Does your amendment change the risk profile of the study or risks/benefits of the study?

- ◯ Yes
- ◉ No

A1.2. Please indicate from the below, any documents that have been amended.

You will be required to re-upload these as part of your re-submission.

- [x] Participant Information Sheet
- [ ] Consent Form
- [x] Questionnaires
- [ ] Debrief
- [ ] Sussex Direct (legacy) application
- [x] Any other uploads
- [ ] None of the above have been amended

A1.2.1. Please describe

Tapping tasks and Heartbeat Counting Task have been included as part of the interoception battery part of the study. The recruitment poster and invitation texts have also been updated to reflect the study duration change.

Applicant

The declarations are to be completed by the applicant, please ensure that you read and sign at the end so your application will be submitted. Failure to do so will result in delays.

5. By submitting your own application, you are agreeing to the following declarations

- [x] I confirm that I have read UoS Code of Practice for Research

  UoS Code of Practice for Research

  Research Governance Standard Operating Procedures (including ethical review)

- [x] The information in this form is accurate to the best of my knowledge and belief, and I take full responsibility for it.
- [x] I understand that I am responsible for monitoring the research at all times and recording any unexpected events.
- [x] If any serious adverse events arise in relation to the research, I understand that I am responsible for immediately stopping the research and alerting the F-REC Chair within 24 hours of the occurrence.
- [x] I am aware of my responsibility to comply with the current requirements of the law and relevant guidelines relating to security and confidentiality of personal data.
- [x] I understand that research records / data may be subject to inspection for audit purposes if required in future.
- [x] I understand that I may not commence this research until I have been notified that the project has ethical approval.
- [x] If there is a substantial change in topic/methodology or risk categorisation I confirm that I will submit an amendment to outline the changes.
- [x] Research records will be held in accordance with the Data Protection Act 2018 as detailed by University Guidance

  Data Protection Act 2018

  University Guidance

- [x] I declare that I have reported all changes for this project throughout the application form
- [x] I will not undertake any of these changes prior to approval

Use the 'Sign' button below to sign your application.

Signature

Signed: This form was signed by . Dominique Makowski (D.Makowski@sussex.ac.uk) on 09/06/2026 4:38 PM

Steps to Request Signature from your Supervisor

Step 1. Give your supervisor access to your application using the 'Roles' Roles button (this will also share them into correspondence emails).

Step 2. Press the 'Request Signature' button below to request your supervisor to sign your application.

You supervisor will review your application and sign the form, and then your form will automatically be submitted for review. You will receive an email confirming once your application has been submitted.

Supervisor Signature

Signed: This form was signed by . Dominique Makowski (D.Makowski@sussex.ac.uk) on 09/06/2026 5:45 PM

For Supervisor: Signature Information

Please ensure that you have read the application in full and discussed any potential changes with the applicant before signing this off for review. Failure to do so will result in delay.

In reviewing and authorising a student’s ethics application, you are confirming that all reasonable steps have been taken to ensure that the project is conducted ethically, provided the student follows what has been agreed. You are also confirming that you will be monitoring the student’s project, as it proceeds.